Project A – Blood Pressure Analysis and Statistical Interpretation

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Objective

This R project analyzes a blood pressure dataset to study how **dose** and **gender** affect **blood pressure reduction**.
The workflow includes descriptive statistics, visualizations, normality and variance testing, confidence interval estimation, hypothesis testing, and linear regression modeling.

Workflow Overview: 1. Load required R packages.
2. Import and inspect the dataset.
3. Perform descriptive and visual statistical analyses.
4. Assess data normality and homogeneity of variance.
5. Conduct hypothesis tests and ANOVA.
6. Build and interpret a linear regression model.

1.0 Load Packages

```
library(googledrive)   # For downloading data directly from Google Drive
library(dplyr)         # For data manipulation (filtering, grouping, summarise)
library(ggplot2)       # For creating plots and visualizations
library(psych)         # For detailed descriptive statistics (describe())
library(car)           # For regression diagnostics and tests (leveneTest, etc.)
library(Hmisc)         # For additional statistical utilities and summaries
```

2.0 Download and Load Dataset

2.1 Access and load dataset

```
drive_deauth()                                                          # Disable Drive auth to acc
file_id <- "1mGrXc0VuNndsqbR6GKhLLv7jkTQ20Ubn"                          # Google Drive file ID for
drive_download(as_id(file_id), path = "BloodPressure.RData", overwrite = TRUE)  # Download dataset to l
load("BloodPressure.RData")                                             # Load RData file into R en
```

3.0 Explore Data

3.1 Inspect structure and summary

```r
str(BloodPressure)    # Show structure: variables, types, and sample values
```

```
## 'data.frame':    40 obs. of  3 variables:
##  $ dose        : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ bp.reduction: num  3 5 -2 -5 0 5 -1 0 -8 -6 ...
##  $ gender      : int  0 0 0 1 0 0 1 1 1 1 ...
```

```r
summary(BloodPressure) # Provide summary statistics (min, 1st Qu., median, mean, etc.)
```

```
##       dose        bp.reduction        gender
##  Min.   : 0.00   Min.   :-8.000   Min.   :0.00
##  1st Qu.: 1.50   1st Qu.: 2.750   1st Qu.:0.00
##  Median : 3.50   Median : 6.500   Median :0.00
##  Mean   : 4.25   Mean   : 7.625   Mean   :0.45
##  3rd Qu.: 6.25   3rd Qu.:13.000   3rd Qu.:1.00
##  Max.   :10.00   Max.   :26.000   Max.   :1.00
```

```r
head(BloodPressure)    # Display first 6 rows for quick inspection
```

```
##   dose bp.reduction gender
## 1    0            3      0
## 2    0            5      0
## 3    0           -2      0
## 4    0           -5      1
## 5    0            0      0
## 6    0            5      0
```

```r
names(BloodPressure)   # List column names for reference
```

```
## [1] "dose"         "bp.reduction" "gender"
```

4.0 Descriptive Statistics

4.1 Basic summaries and correlations

```r
describe(BloodPressure)                                    # Detailed descriptive stats (mean, sd, skew
```

```
## BloodPressure
##
##  3  Variables      40  Observations
## ---------------------------------------------------------------------------
```

```
## dose
##        n  missing distinct     Info     Mean  pMedian      Gmd
##       40        0        4    0.938     4.25     4.25    4.231
##
## Value         0    2    5   10
## Frequency    10   10   10   10
## Proportion 0.25 0.25 0.25 0.25
##
## For the frequency table, variable is rounded to the nearest 0
## ----------------------------------------------------------------------------------
## bp.reduction
##        n  missing distinct     Info     Mean  pMedian      Gmd      .05
##       40        0       23    0.997    7.625      7.5    9.042    -5.05
##      .10      .25      .50      .75      .90      .95
##    -1.10     2.75     6.50    13.00    17.10    21.15
##
## lowest : -8 -6 -5 -2 -1, highest: 17 18 21 24 26
## ----------------------------------------------------------------------------------
## gender
##        n  missing distinct     Info      Sum     Mean
##       40        0        2    0.743       18     0.45
##
## ----------------------------------------------------------------------------------
```

```r
cor(BloodPressure$bp.reduction, BloodPressure$dose)        # Pearson correlation between dose and BP re
```

```
## [1] 0.8594944
```

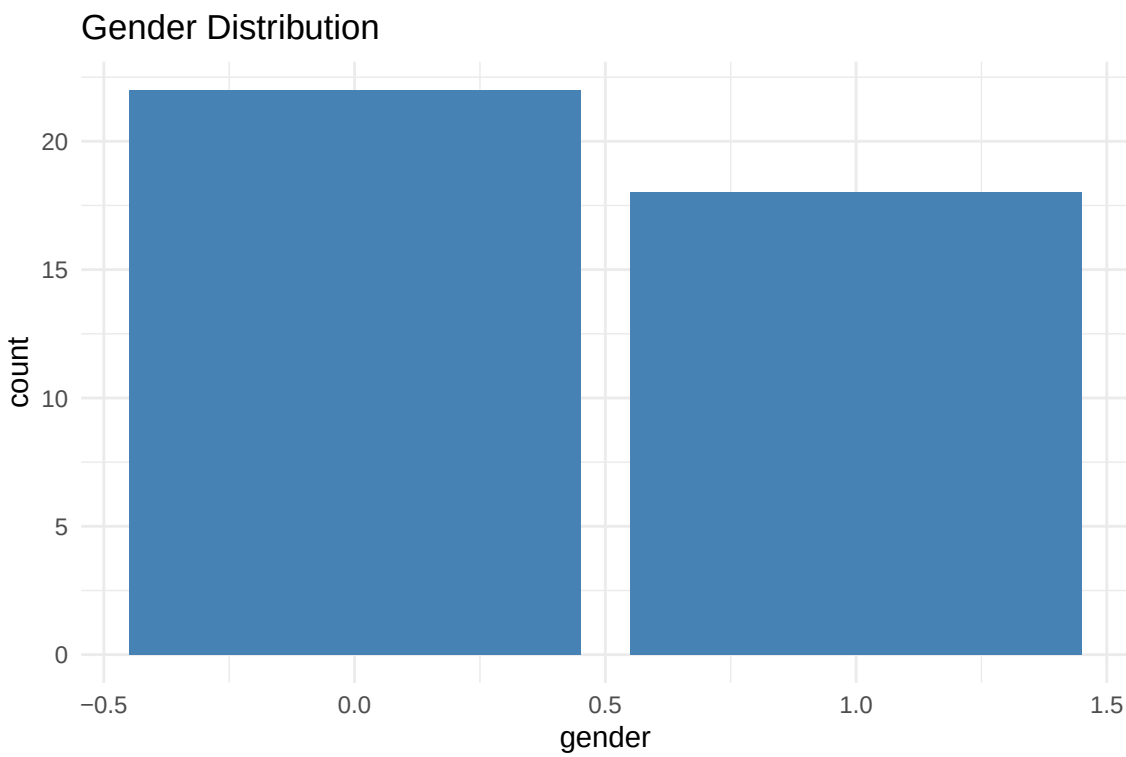
```r
table(BloodPressure$gender)                                # Frequency table for gender variable
```

```
##
##  0  1
## 22 18
```

5.0 Graphics and Visualization

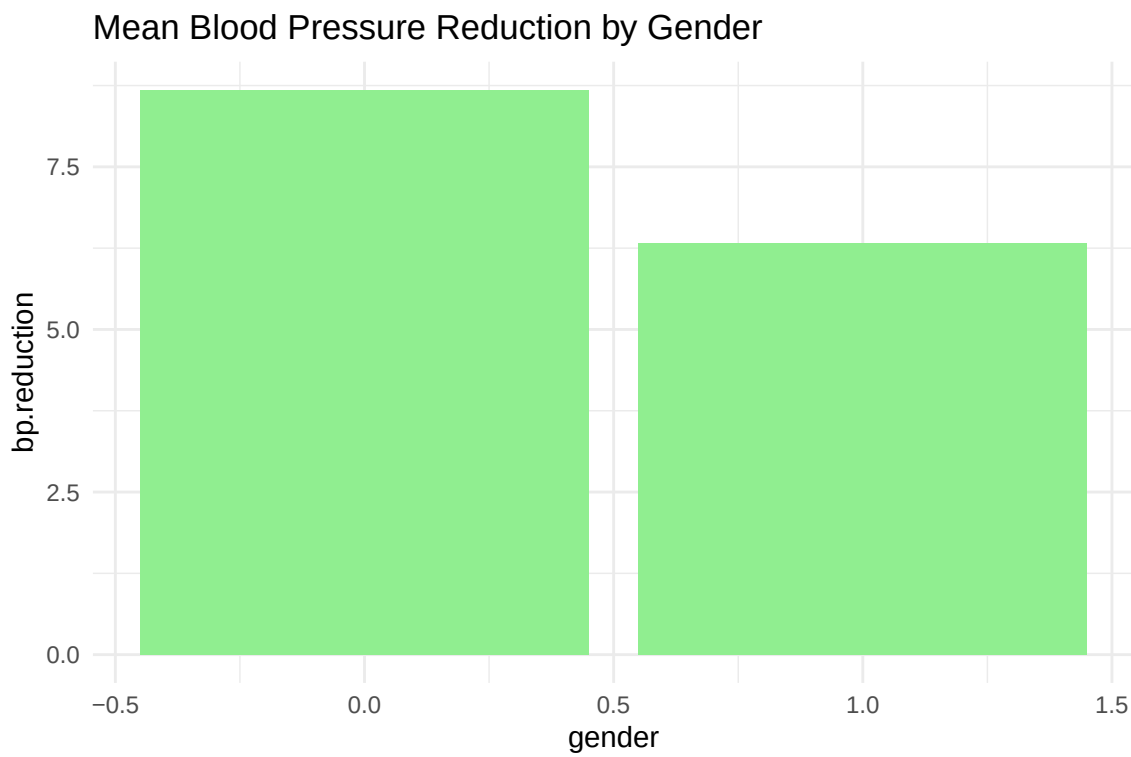
5.1 Bar chart – gender distribution

```r
ggplot(BloodPressure, aes(x = gender)) +                   # Start ggplot mapping gender to x-axis
  geom_bar(fill = "steelblue") +                           # Draw bar chart with blue fill
  labs(title = "Gender Distribution") +                    # Add plot title
  theme_minimal()                                          # Apply a clean minimal theme
```



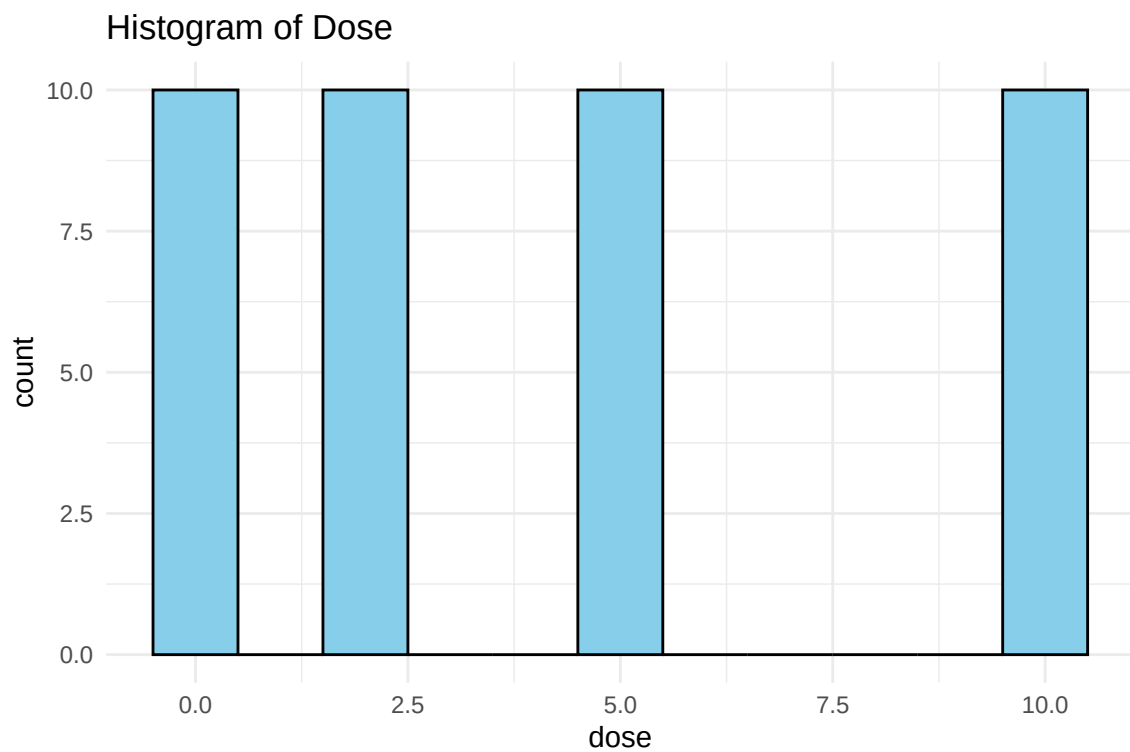
5.2 Bar chart – mean BP reduction by gender

```
ggplot(BloodPressure, aes(x = gender, y = bp.reduction)) + # Map gender to x and mean BP reduction to y
  stat_summary(fun = mean, geom = "bar", fill = "lightgreen") +  # Plot mean as bars with green fill
  labs(title = "Mean Blood Pressure Reduction by Gender") +      # Title explaining the plot
  theme_minimal()                                                 # Minimal theme for clarity
```



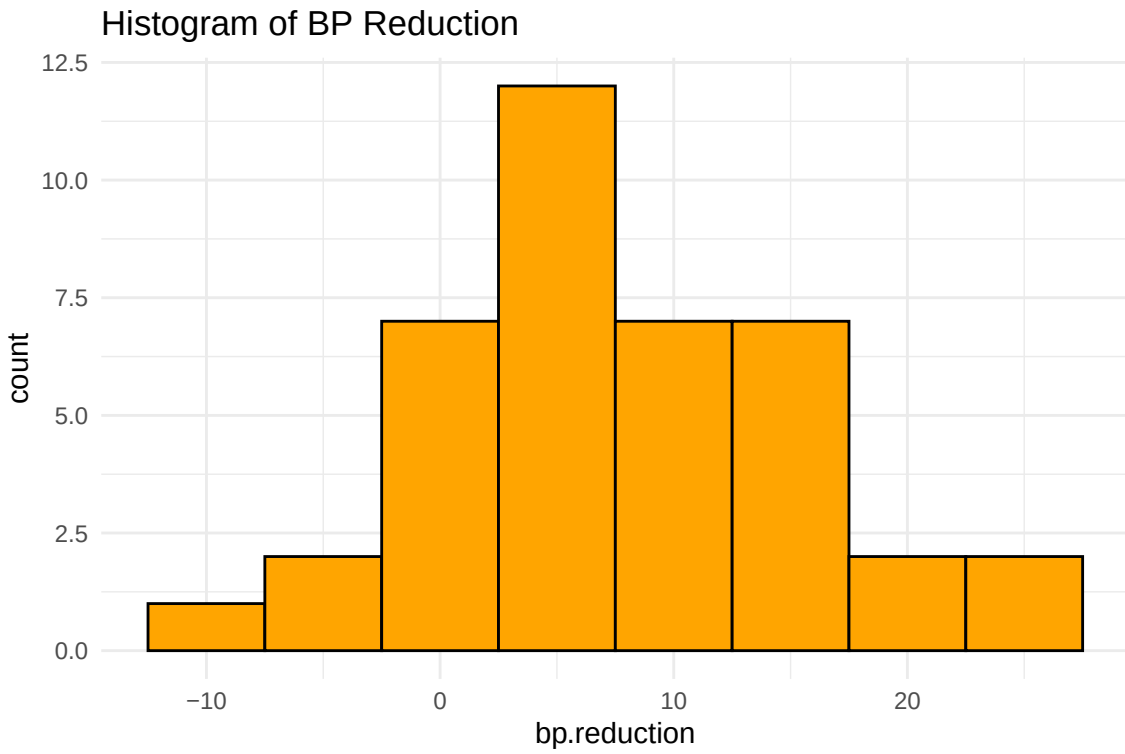
5.3 Histogram – dose distribution

```
ggplot(BloodPressure, aes(x = dose)) +                    # Map dose to x-axis
  geom_histogram(binwidth = 1, fill = "skyblue", color = "black") +  # Histogram with specified bin wid
  labs(title = "Histogram of Dose") +                     # Title for the histogram
  theme_minimal()                                         # Minimal theme for consistency
```



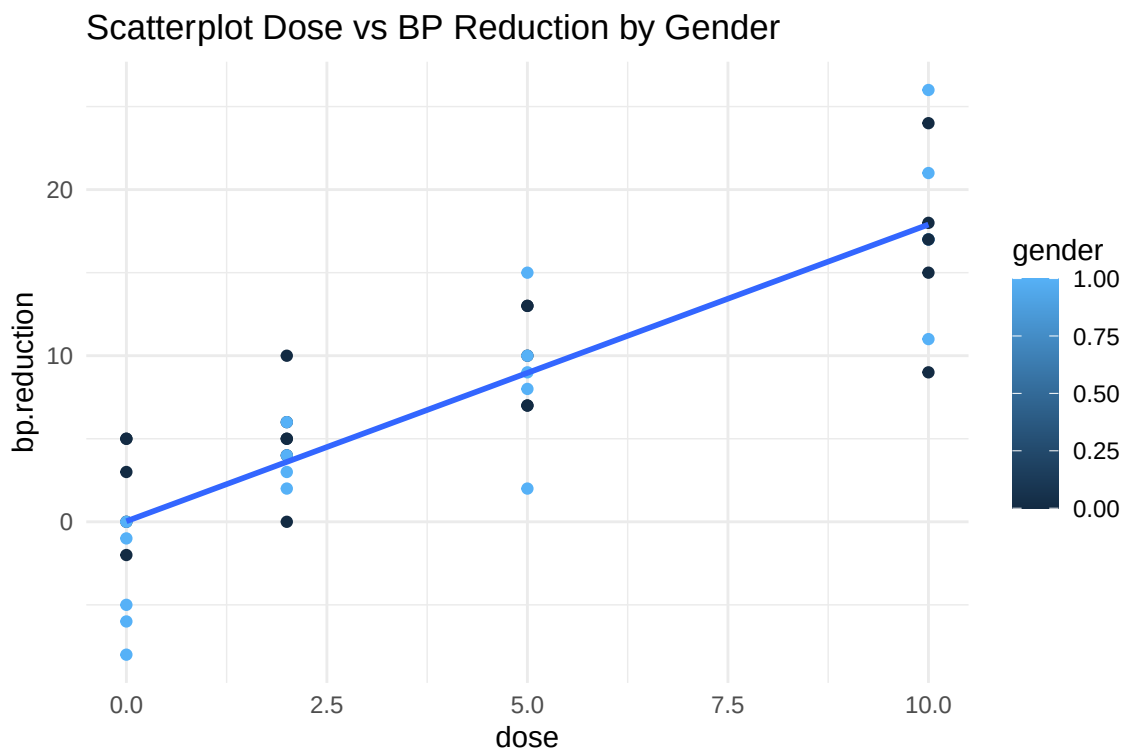
5.4 Histogram – BP reduction distribution

```
ggplot(BloodPressure, aes(x = bp.reduction)) +            # Map BP reduction to x-axis
  geom_histogram(binwidth = 5, fill = "orange", color = "black") +  # Histogram with larger bin width
  labs(title = "Histogram of BP Reduction") +             # Title explaining the distribution plotted
  theme_minimal()                                         # Minimal theme to match other plots
```



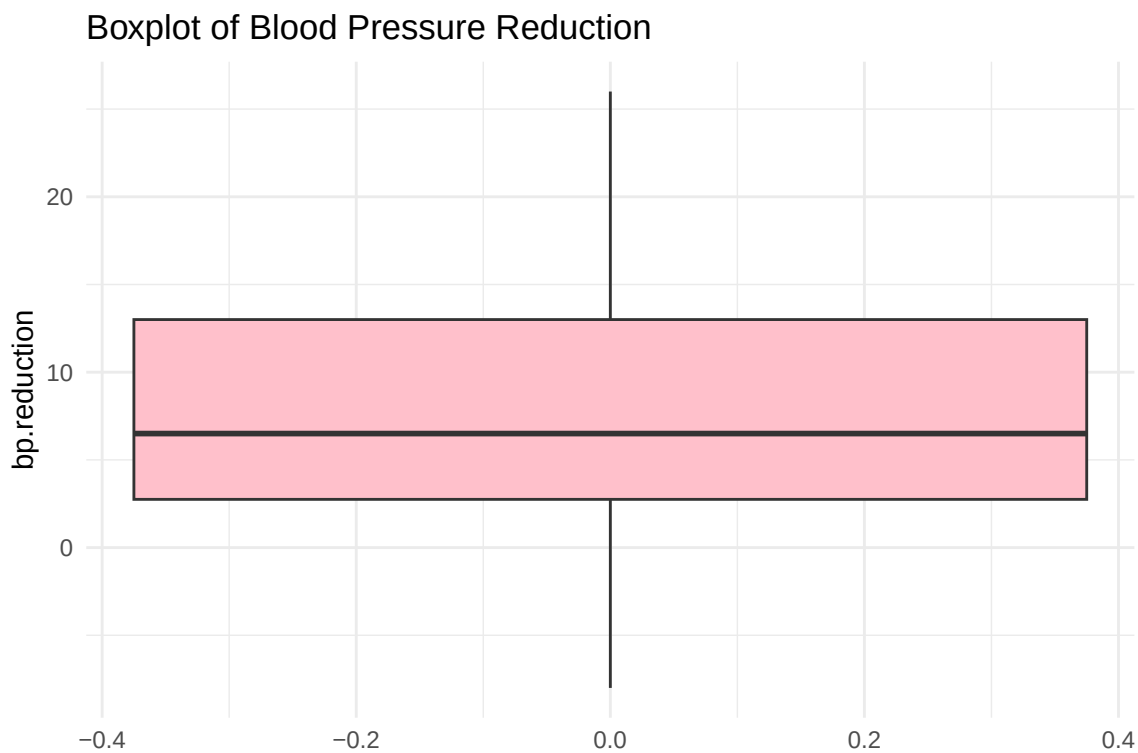
5.5 Scatterplot – dose vs BP reduction by gender

```
ggplot(BloodPressure, aes(x = dose, y = bp.reduction, color = gender)) +  # Map dose to x, BP reduction
  geom_point() +                                                          # Add scatter points
  geom_smooth(method = "lm", se = FALSE) +                                # Add linear regression lines
  labs(title = "Scatterplot Dose vs BP Reduction by Gender") +            # Title for the scatterplot
  theme_minimal()                                                         # Keep visual style consistent
```

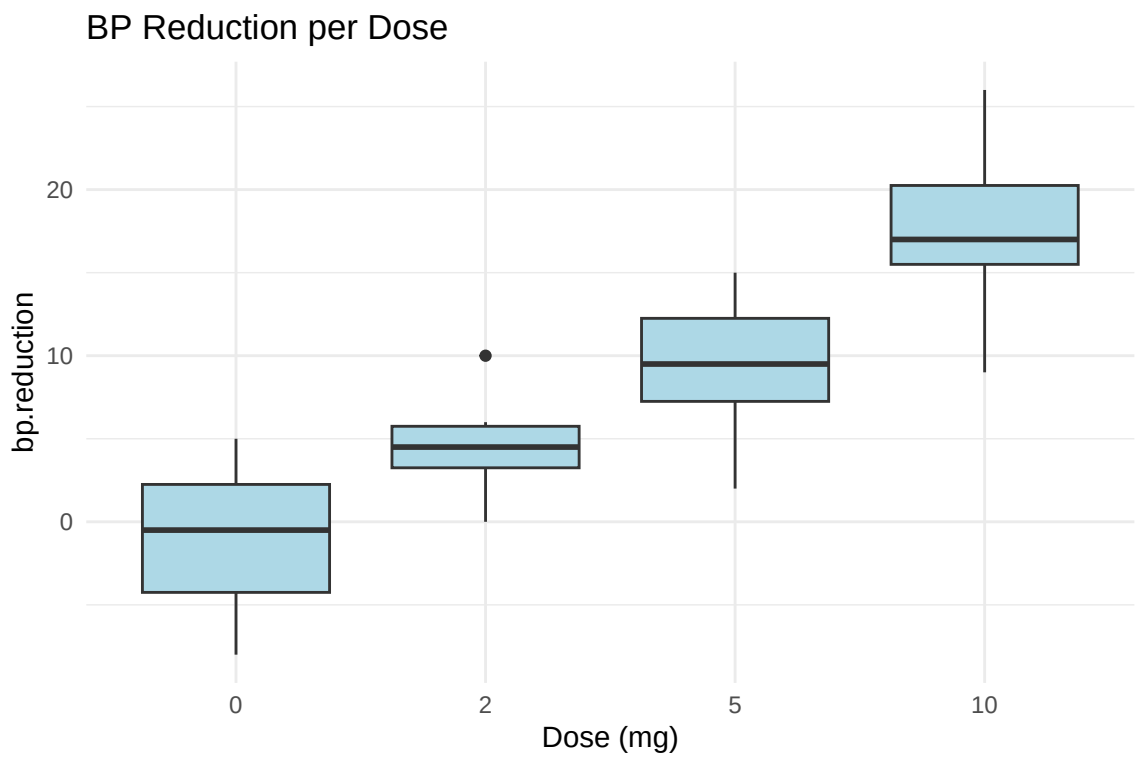


5.6 Boxplots – distribution overview

```
# Boxplot of BP reduction overall
ggplot(BloodPressure, aes(y = bp.reduction)) +          # Map BP reduction to y-axis for boxplot
  geom_boxplot(fill = "pink") +                         # Draw boxplot with pink fill
  labs(title = "Boxplot of Blood Pressure Reduction") + # Title to show purpose of plot
  theme_minimal()                                       # Minimal theme for readability
```



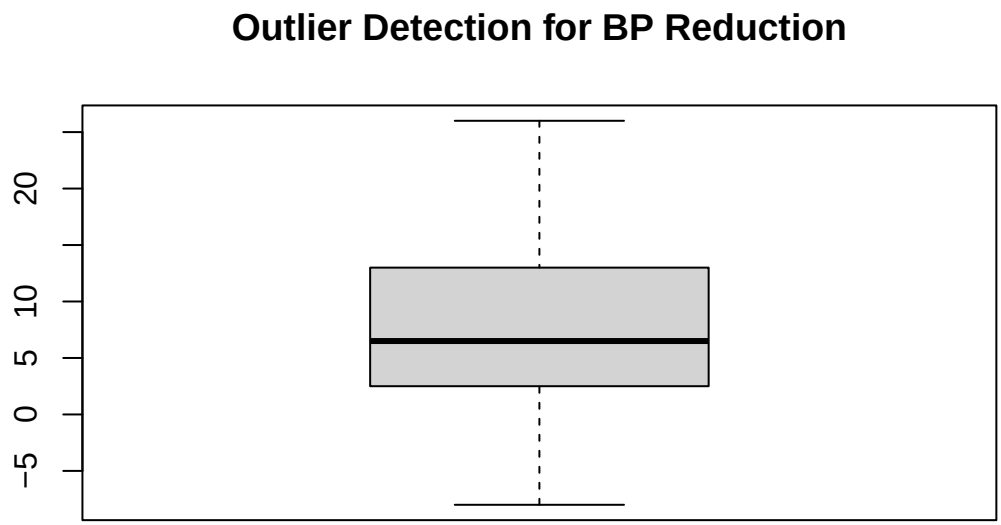

```r
# Boxplot of BP reduction per dose group
ggplot(BloodPressure, aes(x = as.factor(dose), y = bp.reduction)) +  # Treat dose as categorical for gr
  geom_boxplot(fill = "lightblue") +                                  # Fill color for clarity
  labs(title = "BP Reduction per Dose", x = "Dose (mg)") +            # Title and x-label
  theme_minimal()                                                      # Consistent styling
```



6.0 Outlier Detection

6.1 Detect outliers visually

```
boxplot(BloodPressure$bp.reduction, main = "Outlier Detection for BP Reduction")  # Boxplot highlights
```



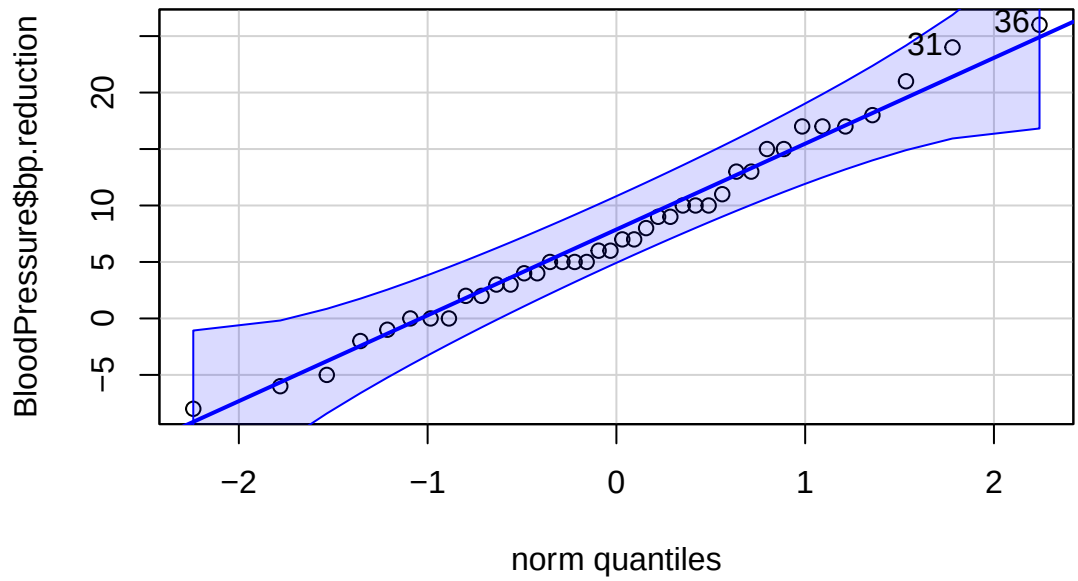
7.0 Normality and Homoscedasticity

7.1 Test for normal distribution

```
shapiro.test(BloodPressure$bp.reduction)          # Shapiro-Wilk test for normality of BP reduction
```

```
##
##  Shapiro-Wilk normality test
##
## data:  BloodPressure$bp.reduction
## W = 0.98321, p-value = 0.806
```

```
qqPlot(BloodPressure$bp.reduction)                # Q-Q plot to visually assess normality (from car pac
```



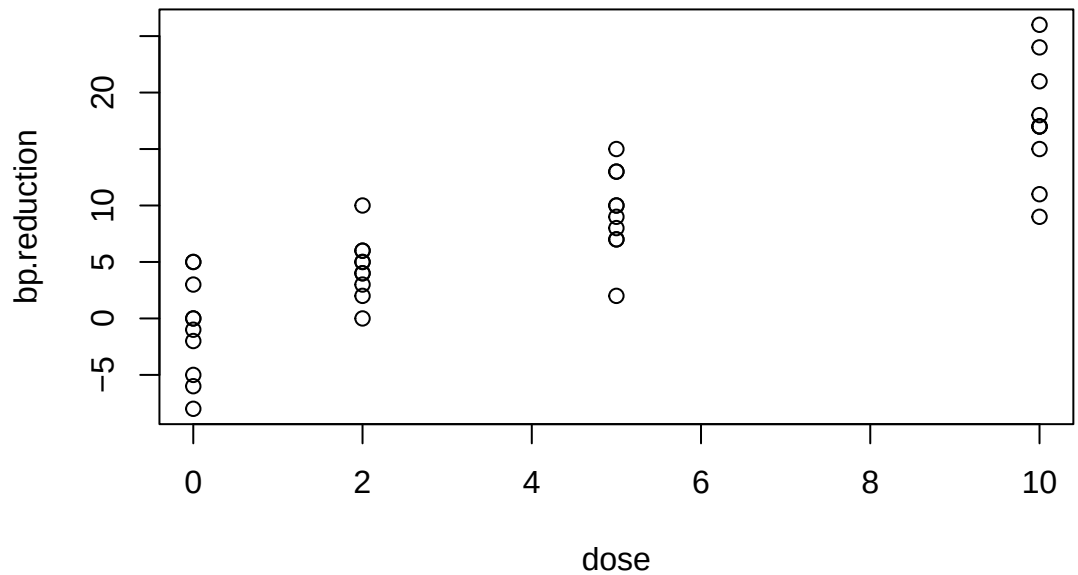

```
## [1] 36 31
```

7.2 Test variance equality (Levene's test)

```r
leveneTest(bp.reduction ~ as.factor(dose), data = BloodPressure)  # Levene's test for homogeneity of va
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  3  0.9517  0.426
##       36
```

```r
plot(bp.reduction ~ dose, data = BloodPressure)                  # Scatter plot to visually inspect var
```



8.0 Confidence Intervals per Dose

8.1 Calculate 90%, 95%, and 99% CI for mean BP reduction

```
BloodPressure %>%
  group_by(dose) %>%                                          # Group data by dose level
  summarise(
    mean_bp = mean(bp.reduction),                             # Compute mean BP reduction within dos

    # 90% Confidence Interval
    ci90_lower = mean_bp - qt(0.95, n()-1) * sd(bp.reduction)/sqrt(n()),  # Lower bound for 90% CI
    ci90_upper = mean_bp + qt(0.95, n()-1) * sd(bp.reduction)/sqrt(n()),  # Upper bound for 90% CI

    # 95% Confidence Interval
    ci95_lower = mean_bp - qt(0.975, n()-1) * sd(bp.reduction)/sqrt(n()), # Lower bound for 95% CI
    ci95_upper = mean_bp + qt(0.975, n()-1) * sd(bp.reduction)/sqrt(n()), # Upper bound for 95% CI

    # 99% Confidence Interval
    ci99_lower = mean_bp - qt(0.995, n()-1) * sd(bp.reduction)/sqrt(n()), # Lower bound for 99% CI
    ci99_upper = mean_bp + qt(0.995, n()-1) * sd(bp.reduction)/sqrt(n())  # Upper bound for 99% CI
  )                                                           # Summarise returns a tibble with mean
```

```
## # A tibble: 4 x 8
##    dose mean_bp ci90_lower ci90_upper ci95_lower ci95_upper ci99_lower
```

```
##   <dbl>   <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1     0    -0.9      -3.50       1.70      -4.11       2.31      -5.51
## 2     2     4.5       2.95       6.05       2.58       6.42       1.75
## 3     5     9.4       7.23      11.6        6.72      12.1        5.55
## 4    10    17.5      14.5       20.5       13.7       21.3       12.1
## # i 1 more variable: ci99_upper <dbl>
```

9.0 Hypothesis Testing

9.1 Compare BP reduction between genders at dose = 0

```
t.test(bp.reduction ~ gender, data = subset(BloodPressure, dose == 0), var.equal = TRUE)  # Independent
```

```
##
##  Two Sample t-test
##
## data:  bp.reduction by gender
## t = 3.011, df = 8, p-value = 0.01679
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##   1.451647 10.948353
## sample estimates:
## mean in group 0 mean in group 1
##             2.2            -4.0
```

9.2 Compare BP reduction between dose 0 and 10 groups

```
t.test(bp.reduction ~ as.factor(dose), data = subset(BloodPressure, dose %in% c(0,10)), var.equal = FALS
```

```
##
##  Welch Two Sample t-test
##
## data:  bp.reduction by as.factor(dose)
## t = -8.4238, df = 17.565, p-value = 1
## alternative hypothesis: true difference in means between group 0 and group 10 is greater than 0
## 95 percent confidence interval:
##  -22.19279       Inf
## sample estimates:
##  mean in group 0 mean in group 10
##             -0.9             17.5
```

9.3 Perform one-way ANOVA across all dose groups

```
anova_model <- aov(bp.reduction ~ as.factor(dose), data = BloodPressure)  # Fit one-way ANOVA model to
summary(anova_model)                                                      # Display ANOVA table (F-stat
```

```
##                 Df Sum Sq Mean Sq F value   Pr(>F)
## as.factor(dose)  3 1831.1   610.4   35.42 7.71e-11 ***
## Residuals       36  620.3    17.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(anova_model)                                                    # Post-hoc Tukey HSD for pair
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = bp.reduction ~ as.factor(dose), data = BloodPressure)
##
## $`as.factor(dose)`
##      diff         lwr       upr     p adj
## 2-0   5.4  0.40037278 10.399627 0.0300758
## 5-0  10.3  5.30037278 15.299627 0.0000161
## 10-0 18.4 13.40037278 23.399627 0.0000000
## 5-2   4.9 -0.09962722  9.899627 0.0564941
## 10-2 13.0  8.00037278 17.999627 0.0000002
## 10-5  8.1  3.10037278 13.099627 0.0005742
```

10.0 Linear Regression Analysis

10.1 Build and interpret linear model

```
BloodPressure$gender <- factor(BloodPressure$gender, levels = c("0","1"), labels = c("male","female"))

lm_model <- lm(bp.reduction ~ dose + gender, data = BloodPressure)   # Fit linear regression predicting
summary(lm_model)                                                    # Show regression coefficients, p-
```

```
##
## Call:
## lm(formula = bp.reduction ~ dose + gender, data = BloodPressure)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.6098 -2.7311  0.1899  2.4279  9.1109
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.8525     1.1480   0.743    0.462
## dose           1.7757     0.1708  10.397 1.57e-12 ***
## genderfemale  -1.7207     1.2932  -1.331    0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.064 on 37 degrees of freedom
## Multiple R-squared:  0.7507, Adjusted R-squared:  0.7372
## F-statistic:  55.7 on 2 and 37 DF,  p-value: 6.928e-12
```

```r
confint(lm_model, level = 0.95)                                    # 95% confidence intervals for coe
```

```
##                  2.5 %    97.5 %
## (Intercept)  -1.473704 3.1786181
## dose          1.429657 2.1218062
## genderfemale -4.340907 0.8995045
```

```r
predict(lm_model, newdata = data.frame(dose = 3, gender = "male"), interval = "confidence")  # Predict
```

```
##        fit      lwr      upr
## 1 6.179651 4.357419 8.001884
```